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Accurate potential energy curve of the LiH^+ molecule calculated with explicitly correlated Gaussian functions

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Very accurate calculations of the ground-state potential energy curve (PEC) of the LiH^+ ion performed with all-electron explicitly correlated Gaussian functions with shifted centers are presented. The variational method is employed. The calculations involve optimization of nonlinear exponential parameters of the Gaussians performed with the aid of the analytical first derivatives of the energy determined with respect to the parameters. The diagonal adiabatic correction is also calculated for each PEC point. The PEC is then used to calculate the vibrational energies of the system. In that calculation, the non-adiabatic effects are accounted for by using an effective vibrational mass obtained by the minimization of the difference between the vibrational energies obtained from the calculations where the Born-Oppenheimer approximation was not assumed and the results of the present calculations. © *2014 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4869517]

I. INTRODUCTION

Radiative association reactions are believed to be the source of LiH and LiH^+ in early universe environment.[1–3] For a correct modeling of the Li chemistry, processes involving LiH^+ need to be considered. Two of these processes which are deemed important are reactions involving LiH^+ with atomic hydrogen[4–6]

$$LiH^+ + H \rightarrow Li^+ + H_2 \quad \text{and} \quad LiH^+ + H \rightarrow Li^+ + H + H. \tag{1}$$

Despite the astrophysical relevance, the spectroscopy of the LiH^+ is still not experimentally elucidated. However, there have been some theoretical calculations concerning the LiH^+ ground-state potential energy curve (PEC) and its rovibrational transitions. The most accurate to date LiH^+ PEC is that of Gadéa and Leininger[7] calculated using a large atomic-orbital basis set and the coupled-cluster method with single, double, and (perturbatively determined) triple excitations (CCSD(T)). We make a comparison of the present results with the results obtained by Gadéa and Leininger[7] in Sec. III in this work.

The present work is a continuation of our accurate Born-Oppenheimer (BO) calculations for small diatomic systems[8,9,11,12] performed with large basis sets of all-electron explicitly correlated Gaussian functions with shifted centers (also called floating explicitly correlated Gaussians (ECGs)). In these calculations, we have generated BO potential energy curves and calculated rovibrational transitions. 2400 ECGs were used to expand the electronic wave functions for four-electron molecules, 1200 ECGs for three-electron molecules, and 600 ECGs for two-electron molecular systems. These many functions ensured that the accuracy of the PECs was sufficient to very accurately calculate the energies of the rovibrational states of the molecules by solving the one-dimensional nuclear-motion Schrödinger equation where the PECs provide the interaction potential.[13] With the high accuracy achieved for these energies and for the corresponding rovibrational transition frequencies it was possible to conclude that the inaccuracies due to neglecting the leading contributions from the relativistic, quantum electrodynamics (QED), and nonadiabatic effects are larger than the inaccuracies of the adiabatic PECs. Thus, instead of improving the accuracy of the PECs, an account for these omitted effects should be included in the calculations. With that one can further improve the agreement between the theoretical and experimental results.

The larger is the molecule (in terms of the number of electrons and the number of nuclei) the larger computation effort is involved in the calculations performed with all-electron ECGs to achieve a consistently high level of accuracy. This effort goes primarily into the optimization of the nonlinear parameters of the basis set. When a system with a larger number of electrons is considered, not only the ECG basis set has to be larger, but also the integrals that need to be calculated have higher dimensionality ($3n$) and the number of components resulting from the implementation of the permutational symmetry in each integral increases (as $n!$). In view of the rapidly increasing cost of the PEC calculation, it becomes important to be able to estimate how accurate the PEC needs to be and what minimum size of the ECG basis set needs to be employed in the PEC calculation to obtain the rovibrational transitions with the desired level of accuracy. Thus, one of the main aims of the present calculations, besides generating an accurate PEC for the LiH^+ molecule, is to investigate the relation between the accuracy of the PEC and the rovibrational transitions obtained by solving the nuclear-motion Schrödinger equation with that PEC.

II. THE PEC CALCULATIONS

A. The wave function

As already showed in the early work of Hylleraas[14] on the helium atom, to accurately describe the interacting

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electrons in a molecule, the wave function needs to explicitly depend on the inter-electronic distances. This conclusion and the realization that the computational difficulties associated with the use of the Hylleraas functions for systems with more than two-three electrons are not going to be easy to overcome, motivated Boys[15] and Singer[16] to introduce Gaussian basis functions that explicitly depend on the inter-electron distances in the exponential factor. These functions are now often referred to as explicitly correlated (or exponentially correlated) Gaussian functions (ECGs). ECGs, due to the simplicity of calculating the Hamiltonian matrix elements with those functions, have become popular in very accurate quantum mechanical calculations of bound states of small atoms and molecules. They have been successfully applied in very accurate calculations of small atoms and molecules performed with and without the assumption of the BO approximation.[17] For example, they have been recently used in the calculations of the most complete BO PES of the H_3^+ ion,[18] the BO PEC for the He_2^+ ion,[19] and the $(H_2)_2$ four-electron cluster.[8]

In order to achieve high accuracy in both BO and non-BO ECG calculations, one needs to perform a thorough optimization of the Gaussian nonlinear parameters. In our works, this has been done with the aid of the analytic energy gradient determined with respect to these parameters. Also, it is crucial in a ECG calculation to develop a procedure for a systematic increase of the basis-set size (i.e., to grow the ECG basis from a small size to a size that assures the desired accuracy of the results). In recent years, we have developed and implemented several methods for more efficient generation of PESes of small molecular systems employing ECGs.[20] As computational resources become increasingly more accessible and affordable, such calculations become more feasible. However, as the size of the molecule in terms of the number of electrons and the number of nuclei increases, the complexity and the cost of the ECG PES calculations also increases. Let us take two-electron systems with different numbers of nuclei as examples. For a system with one nucleus, e.g., the helium atom, one can reach 1×10^{-9} Hartree (Eh hereafter) accuracy of the energy with only 200 Gaussians.[21] In case of two and three nuclei, H_2 and H_3^+, 500 and 1000 Gaussians are needed, respectively, to achieve similar accuracy.[18,22] Going to systems with more than two electrons, the four-electron, four-nuclear $(H_2)_2$ cluster requires at least 7000 ECGs, and even with 7000 ECGs the accuracy reached is only 1×10^{-6} Eh.[8]

Another issue that arises in the optimization of the ECG nonlinear parameters are linear dependencies between basis functions. They appear because the optimization procedure may try to describe certain features of the wave function by pairs of very similar ECGs with coefficients of nearly the same magnitude, but opposite signs, instead of single ECGs, which would provide equally good description of the features. Linear dependencies are undesirable because they may cause loss of the numerical accuracy in the calculation. To effectively deal with the linear-dependency problem a rather sophisticated procedure is required.[9]

The spatial part of the electronic wave function representing the ground state of the LiH^+ ion in the present calculations is a linear combination of ECGs, $\phi_k(\mathbf{r})$

$$\Psi(\mathbf{r}) = \sum_{k=1}^{M} c_k \phi_k(\mathbf{r}), \tag{2}$$

where M is the size of the basis set and c_k are the linear expansion coefficients. The ϕ_k's are the following correlated Gaussians with shifted centers:

$$\phi_k(\mathbf{r}) = \exp[-(\mathbf{r} - \mathbf{s}_k)'(\mathbf{A}_k \otimes \mathbf{I}_3)(\mathbf{r} - \mathbf{s}_k)]. \tag{3}$$

In Eq. (3), the symbol $\otimes$ denotes the Kronecker product, $\mathbf{I}_3$ is a 3×3 identity matrix, $\mathbf{r}$ is the $3n$ vector of the electron Cartesian coordinates, $\mathbf{s}_k$ is a $3n$-dimensional shift vector of the Gaussian centers, n is the number of electrons, and $\mathbf{A}_k$ is a symmetric matrix. For the ECG to be square integrable $\mathbf{A}_k$ has to be a positive definite matrix.[23] This automatically happens if $\mathbf{A}_k$ is represented in the Cholesky-factorized form as: $\mathbf{A}_k = \mathbf{L}_k\mathbf{L}_k'$, where $\mathbf{L}_k$ is a lower triangular matrix and $\mathbf{L}_k'$ is its transpose.

The total electronic wave function that includes spin must be antisymmetric with respect to the permutation of the electron labels. Thus, an appropriate symmetry projection need is applied to the wave function to assure that. In this work, we use the spin-free formalism. Thus, the Young projection operator, $\hat{Y}$, which is constructed to impose the proper permutation symmetry, is only applied to the spatial part of the wave function. $\hat{Y}$ is a linear combination of permutational operators, $\hat{P}_\gamma$, and, as the Hamiltonian is invariant with respect to all permutations of the electronic labels, in the calculation of the overlap and Hamiltonian matrix elements these operators are applied to the *ket* only. In brief, the *ket* basis functions in those matrix elements are operated on with the permutation operator $\hat{P} = \hat{Y}^\dagger\hat{Y}$ (the dagger stands for conjugate). The $\hat{Y}$ operator can be derived using the appropriate Young tableaux representing the doublet ground state of LiH^+. For this state, the Young operator can be chosen as: $\hat{Y} = (\hat{1} + \hat{P}_{23})(\hat{1} - \hat{P}_{12})$, where $\hat{1}$ is the identity operator and $\hat{P}_{ij}$ is the permutation of the ith and jth electrons.

The first step of the present calculations involves building a LiH^+ ECG basis set at the equilibrium internuclear distance for this ion. The target number of ECGs is set to be 1200. The procedure for building an ECG basis set was described previously[11] and it is applied here. In the building process, smaller-size basis sets are also generated. They allow for examining the energy convergence with the basis set size. This convergence is shown in Table I for some selected PEC points. For each point, the convergence of the diagonal adiabatic correction is also shown. For the equilibrium internuclear distance of 4.143 a_0, we also show in Table I the LiH^+ total energy reported by Gadéa and Leininger[7] obtained with the CCSD(T) method and large orbital basis set. This result is by 0.0005 E_h higher than our result. Even though the CCSD(T) energy is not strictly variational, for a three-electron system it should be close to the energy obtained in the same basis set with the full configuration-interaction (FCI) method.

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TABLE I. Convergence of the BO energy and the adiabatic correction of $^7LiH^+$ with the basis set size at selected internuclear distances. The adiabatic correction is split into the contributions from the nuclei of ^{7}Li and H. All quantities are in E_h.

	Internuclear distance (a_0)											
	3.5			4.143			5			15		
Basis size (M)	BO energy	^{7}Li	H	BO energy	^{7}Li	H	BO energy	^{7}Li	H	BO energy	^{7}Li	H
75	−7.78249952	0.00059213	0.00025881	−7.78473159	0.00059180	0.00025874	−7.78350941	0.00059173	0.00026270	−7.77986710	0.00059195	0.00027175
150	−7.78281503	0.00059267	0.00026062	−7.78502865	0.00059212	0.00026108	−7.78375623	0.00059186	0.00026443	−7.77994538	0.00059194	0.00027201
300	−7.78284467	0.00059290	0.00026133	−7.78504270	0.00059227	0.00026166	−7.78376115	0.00059192	0.00026515	−7.77995239	0.00059194	0.00027207
600	−7.78285982	0.00059324	0.00026188	−7.78506353	0.00059249	0.00026229	−7.78377278	0.00059211	0.00026569	−7.77995870	0.00059194	0.00027219
1200	−7.78286035	0.00059330	0.00026199	−7.78506392	0.00059253	0.00026238	−7.78377308	0.00059215	0.00026578	−7.77995893	0.00059195	0.00027223
Ref. 7				−7.784539								

B. Adiabatic correction

The adiabatic correction is calculated using the following Born-Handy formula:[24]

$$E_{ad} = \langle\Psi| - \frac{1}{2}\sum_{\alpha=1}^{K}\frac{1}{M_\alpha}\sum_{i_\alpha=1}^{3}\frac{\partial^2}{\partial Q_{i_\alpha}^2}|\Psi\rangle, \tag{4}$$

where M_α is the mass of nucleus α (12 786.393m_e for the ^{7}Li nucleus and 1836.15267221m_e for the proton, where $m_e = 1$ is the electron mass), Q_{i_α} is the ith coordinate of the nucleus α, and K is the number of nuclei in the system. In Eq. (4), instead of directly differentiating the electronic wave function with respect to the nuclear coordinate Q_{i_α}, we follow the approach proposed by Cencek and Kutzelnigg[25] to approximate the derivative numerically as

$$\frac{\partial\Psi}{\partial Q_{i_\alpha}} \simeq \frac{\Psi\left(Q_{i_\alpha} + \frac{1}{2}\Delta Q_{i_\alpha}\right) - \Psi\left(Q_{i_\alpha} - \frac{1}{2}\Delta Q_{i_\alpha}\right)}{\Delta Q_{i_\alpha}}. \tag{5}$$

The Gaussian centers for the wave functions at $Q_{i_\alpha} \pm \frac{1}{2}\Delta Q_{i_\alpha}$, $\Psi(Q_{i_\alpha} \pm \frac{1}{2}\Delta Q_{i_\alpha})$, are determined from their values in $\Psi(Q_{i_\alpha})$ by the procedure described in Refs. 9 and 25. The $\mathbf{L}_k$ parameters are the same in $\Psi(Q_{i_\alpha} \pm \frac{1}{2}\Delta Q_{i_\alpha})$ as they are in $\Psi(Q_{i_\alpha})$, and the linear expansion coefficients are recalculated using the variational method. For $\Delta Q_{i_\alpha} = 1 \times 10^{-4}$ such an approach gives sufficient accuracy in the calculation of the adiabatic correction.[9] As the adiabatic correction for LiH^+ can be split into contributions from the Li and H atoms (see Eq. (4)), they can be calculated and shown separately.

III. THE RESULTS

In our previous calculations on three-electron molecular system, the He_2^+ ion,[11] 1200-ECGs basis set was sufficiently large to achieve satisfactory convergence of the BO energy at each PEC point. Though the absolute accuracy of the energies calculated with 1200 ECGs basis set is not exactly established, one can expect it to be better than the difference in the BO energies obtained with 600 and 1200 ECGs of about 3.9×10^{-7} E_h. If the adiabatic PECs generated with basis sets with different sizes were perfectly parallel, the PECs would have the same relative accuracy and the rovibrational transitions obtained using them in the nuclear rovibrational equation would be the same. However, this is not the case for the LiH^+ results obtained with the different sizes of the ECG basis sets. Examination of the results in Table I, shows that the energies converge faster for larger internuclear distances than for the shorter ones. Also, as it can be seen in Table I, the adiabatic corrections (adiabatic correction is split in the table into the contributions from the ^{7}Li and H nuclei) converge faster than the BO energies. Both trends cause the accuracy of the different BO PECs to be somewhat uneven. The complete LiH^+ BO PEC and the adiabatic corrections obtained with 1200 ECGs are shown in the supplementary material.[10] Although the absolute accuracy of the LiH^+ PEC calculated with 1200 ECGs at and near the equilibrium bond length cannot be directly determined, the accuracy of the long-range behavior of the PEC can be tested by comparing it with a PEC obtained by adding the charge-induced-dipole interaction

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TABLE II. Asymptotic behavior of the LiH^+ BO PEC. A comparison of the BO PEC calculated with 1200 ECGs with the charge-induce-dipole interaction determined using the polarizability and hyperpolarizability of the hydrogen atom.[a] The total energies are given in E_h.

Internuclear distance, R (a_0)	Energy estimated using[a]	Energy calculated using 1200 ECGs	Δ (cm^{-1})
20	−7.77992834	−7.77992759	0.165
30	−7.77991627	−7.77991613	0.030
40	−7.77991431	−7.77991421	0.021
50	−7.77991378	−7.77991368	0.021
145	−7.77991342	−7.77991326	0.035

[a]The energy is estimated as: $E(Li^+)$[26]+E(H)+$V(R)$, where $E(Li^+)$ and E(H) are the BO total energies of the Li^+ ion and the hydrogen atom, respectively, $V(R) = -\frac{\alpha_{ZZ}}{2R^4} - \frac{\gamma_{ZZZZ}}{24R^6}$ is the charge-induced-dipole interaction energy, $\alpha_{ZZ} = 4.5$ a.u.,[27] $\gamma_{ZZZZ} = 1333.125$ a.u.,[28] and R is the internuclear distance.

energy to the energies of the Li^+ ion and the hydrogen atom. This comparison is shown in Table II. As one can see, for the $R \geq 30$ a_0 points the two PECs are almost parallel to each other. In the next step, we evaluate how the overall error in the adiabatic PEC influences the accuracy of the vibrational transitions calculated with the PEC.

The ultimate test of the accuracy of the theoretical results is their comparison with the high-quality experimental data, provided that such data are available. As the rovibrational spectroscopic transitions for the LiH^+ ion are still unknown experimentally, the vibrational transitions calculated in this work using the PEC obtained by summing the BO energies and the adiabatic corrections are compared with the very accurate nonadiabatic nonrelativistic calculation performed for pure vibrational states of LiH^+ without assuming the BO approximation.[29,30] Such a comparison of the present results is even more appropriate than the comparison with the experiment, because all the effects, adiabatic and nonadiabatic, which are fully accounted for in non-BO calculations, are either fully accounted for (adiabatic effects) or partially accounted (nonadiabatic effects) for in the present calculations. While the adiabatic corrections are directly included in the adiabatic PEC we calculate, the nonadiabatic effects are partially included using an effective vibrational reduced mass. This reduced-mass value is obtained by the minimization of the difference between the vibrational transitions between adjacent states obtained in this work ($E_{ad}^{\nu'\to\nu'-1}$) and the transitions obtained in the non-BO calculations ($E_{non-BO}^{\nu'\to\nu'-1}$)[29,30] for the five lowest vibrational states of LiH^+

$$\min\left[\sum_{\nu'=1}^{5}\left(E_{ad}^{\nu'\to\nu'-1} - E_{non-BO}^{\nu'\to\nu'-1}\right)^2\right]. \quad (6)$$

For the vibrational calculations[13] based on the BO approximation to be even more precise, the effective vibrational reduced mass should vary with the internuclear distance.[31,32] However, such reduce-mass variability is not included in the present work. The optimization of the effective vibrational reduced mass using Eq. (6) allows to effectively account for the nonadiabatic effects in the calculation of the vibrational states. The effective reduced mass is only optimized using

TABLE III. Convergence of $[\sum_{\nu'=1}^{5}|E_{ad}^{\nu'\to\nu'-1} - E_{non-BO}^{\nu'\to\nu'-1}|]$ for the pure vibrational transitions of $^7LiH^+$ with the basis set size. The reduced mass equal to $\mu_{vib} = 0.881234$ amu is used. All quantities are in cm^{-1}. The $E_{non-BO}^{\nu'\to\nu'-1}$ is from Ref. 29.

Basis size	$\left[\sum_{\nu'=1}^{5}\left\|E_{ad}^{\nu'\to\nu'-1} - E_{non-BO}^{\nu'\to\nu'-1}\right\|\right]$
75	46.604
150	5.570
300	3.679
600	0.112×10^{-3}
1200	0.107×10^{-3}

the results obtained with the 1200-ECGs PEC and the value 0.881234 amu is obtained. This value is then used to perform vibrational calculations with the PECs obtained with the other basis sets. A comparison of the results for the quantity defined by Eq. (6) is shown in Table III. As it can be seen and as expected, the larger the basis set the more accurate the results.

It is interesting to compare the effective vibrational reduced mass of 0.881234 amu with the reduced mass of a lithium atom and a proton ($Li + H^+$) and the reduced mass of a lithium plus ion and a hydrogen atom ($Li^+ + H$). The latter two values are 0.880819 and 0.881230 amu, respectively. As one can see the effective reduced mass is almost identical with the ($Li^+ + H$) reduced mass. This can be expected because at all internuclear separations like the LiH^+ ion the electron hole stays at the lithium atom and the electronic structure of the system to a very good approximation can be described as $Li^+ + H$. One can calculate the vibrational frequencies using the ($Li^+ + H$) and ($Li + H^+$) reduced masses. The results of such calculations are shown in Table IV. As one notices, the results obtained for the ($Li^+ + H$) reduced mass are only slightly less accurate from the non-BO results than the results obtained with the effective reduced mass. However, the results obtained with the $Li + H^+$ reduced mass are noticeably (by a factor of two) less accurate. The calculation clearly demonstrates the sensitivity of the calculation of the vibrational frequencies to the choice of the reduced mass and shows that this mass has to be chosen to reflect the mass (charge) distribution in the system. In case of the system changing the character of its bond with the internuclear separation like LiH, which near the equilibrium is more accurately described as $Li^+ + H^-$ and at larger internuclear distances as $Li + H$, one should use a variable reduced mass dependent on the internuclear distance reflecting the change of the nature of the bond from ionic to covalent. The variable-mass approach is clearly not necessary for LiH^+.

If one evaluates the vibrational transitions level by level, as shown in Table IV, one observes that the accuracy achieved with 600 ECGs is only slightly worse than that achieved with 1200 ECGs. In both cases, the accuracy is better than 0.03 cm^{-1}. This suggests that it is probably not possible to improve the accuracy of the adiabatic PEC beyond 1×10^{-7} E_h without including the leading relativistic corrections. By comparing the results in Tables I and IV, one can conclude that to achieve 0.03 cm^{-1} accuracy for the vibrational transitions the

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TABLE IV. Convergence of the pure vibrational transitions of $^7LiH^+$ with respect to the basis set size. $E_{ad} - E_{non-BO}$ is the difference between the value obtained in the present calculation and the value obtained in the calculation where the BO approximation was not assumed.[29] The effective vibrational reduced mass of $\mu_{vib} = 0.881234$ amu, reduced mass of $\mu_{Li+H^+} = 0.880819$ amu, and reduced mass of $\mu_{Li^+ +H} = 0.881230$ amu are used to obtain E_{ad}. All quantities are in cm^{-1}. The E_{non-BO} is from Ref. 29.

			$E_{ad} - E_{non-BO}$		
$\nu' \to \nu''$	Basis size	Transition frequency	μ_{vib}	μ_{Li+H^+}	$\mu_{Li^+ +H}$
$1 \to 0$	75	348.8432	−6.168	…	…
	150	355.2766	0.265	…	…
	300	357.3418	2.330	…	…
	600	355.0057	-5.781×10^{-3}	…	…
	1200	355.0115	-3.296×10^{-5}	6.188×10^{-2}	5.637×10^{-4}
$2 \to 1$	75	254.1321	−7.633	…	…
	150	260.9878	−0.778	…	…
	300	262.3865	0.621	…	…
	600	261.7888	2.327×10^{-2}	…	…
	1200	261.7937	2.824×10^{-2}	4.592×10^{-2}	2.841×10^{-2}
$3 \to 2$	75	156.6951	−13.285	…	…
	150	168.4405	−1.540	…	…
	300	169.7769	−0.203	…	…
	600	169.9562	-2.399×10^{-2}	…	…
	1200	169.9595	-2.063×10^{-2}	-4.279×10^{-2}	-2.091×10^{-2}
$4 \to 3$	75	77.4477	−12.371	…	…
	150	88.6752	−1.144	…	…
	300	89.5942	−0.225	…	…
	600	89.7846	-3.457×10^{-2}	…	…
	1200	89.7863	-3.286×10^{-2}	-7.642×10^{-2}	-3.328×10^{-2}
$5 \to 4$	75	28.1132	−7.146	…	…
	150	34.4152	−0.844	…	…
	300	34.9597	−0.299	…	…
	600	35.2346	-2.456×10^{-2}	…	…
	1200	35.2343	-2.489×10^{-2}	-6.253×10^{-2}	-2.525×10^{-2}
$6 \to 5$	75	6.5288	…	…	…
	150	8.3917	…	…	…
	300	8.6838	…	…	…
	600	9.4230	…	…	…
	1200	8.7707	…	…	…

PEC needs to be calculated with the accuracy better than 1×10^{-7} E_h. However, to achieve only a sub-wavenumber accuracy, the PEC needs to be calculated with the accuracy of about 1×10^{-5} E_h.

We consider our transition energies to have accuracy of about 0.03 cm^{-1} within the BO model. Thus, it is interesting to compare them with results obtained by others. There are two sets of results in the literature one by Gadéa and Leininger[7] and the other by Lundsgaard and Rudolph[33] which seems to be the most accurate available. Both were obtained using large orbital basis sets. In the former, the CCSD(T) method was used and in the latter the FCI method. As the two approaches are less accurate than the method employed in the present calculations, some differences are expected to appear between the three sets of results. One can assess these differences by examining the comparison presented in Table V. As one can see the results of Gadéa and Leininger[7] are noticeably closer to our results than the results of Lundsgaard and Rudolph.[33] But the former results are still on average more than 1 cm^{-1} off per transition from our results. This seems to be a difference one can expect to see between frequencies obtained using a PEC from orbital-based calculations and calculations performed with a PEC from all-electron explicitly correlated calculations.

TABLE V. Comparison of the transition energies (in cm^{-1}) for adjacent vibrational states of LiH^+ with the results of Gadéa and Leininger[7] and Lundsgaard and Rudolph.[33]

$\nu' \to \nu''$	Lundsgaard and Rudolph[33]	Gadéa and Leininger[7]	Present results
$1 \to 0$	351.6	353.9	355.01
$2 \to 1$	257.2	260.3	261.79
$3 \to 2$	163.5	168.9	169.96
$4 \to 3$	84.1	88.3	87.79
$5 \to 4$	31.8	34.7	35.23
$6 \to 5$	7.3	9.1	8.77

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IV. SUMMARY

A PEC for the ground state of the LiH^+ ion is calculated using 1200 all-electron ECGs with shifted center. The energy for each PEC point includes the BO nonrelativistic energy and the diagonal adiabatic correction. The nonadiabatic effects are accounted for at the level of solving the vibrational problem with the PEC by using an effective vibrational reduced mass. This mass is obtained by the minimization of the difference between the vibrational frequencies obtained in the present calculations and those obtained in the previous non-BO calculations. It is found that the optimal reduced mass obtained from the minimization is very close to the reduced mass one gets using the Li^+ and H atomic masses. This is an expected result, as the LiH^+ ion be described as Li^+ interacting with a hydrogen atom at all PEC points.

It is determined that 1200 ECGs is optimal basis set for the LiH^+ PEC calculation. Further improvement of the PEC cannot be achieved by increasing the size of the basis, but by including relativistic and QED corrections. Such calculations employing all-electron ECG basis sets will become possible in near future with the software which is under development in our laboratory.

As PECs for more diatomic systems are generated – these include the PEC calculated in this work – we learn more what basis size is sufficient to generate an accurate PEC for a system with a particular number of electrons. This is an important information because using a basis set which is too large may lead to a significant increase of the computational effort, but employing a basis set which is too small may lead to an undesired loss of accuracy in the determination of the vibrational transition frequencies. However, it should be noted that, due to the factorial dependency of the computation time of the variational PEC calculations with all-electron ECGs, these types of calculations will remain limited for the foreseeable future to small molecular systems.

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